Daily Combinatorics Drill #3

Section	Topic	Questions	Target time
A	Rapid Recognition	10	2 min 30 sec
B	Manipulation Drills	10	8 min
C	Substitution & Structure	8	10 min
D	Challenge	5	15 min

New toolkit today: Blocks & Gaps · Circular Arrangements · Repeated Letters · Forced-Order Tricks.

Attempt each question before checking answers. Time yourself. No calculators.

Section A Rapid Recognition ($\leq$15 s each)

Rows, circles, repeated letters. One formula each. Pure recognition.

1. In how many ways can 5 people stand in a row?

2. How many arrangements are there of the letters of LEMON?

3. How many arrangements are there of the letters of BANANA?

4. In how many ways can 4 people sit around a round table? (Rotations count as the same.)

5. Six people stand in a row with Ali first and Zara last. How many arrangements?

6. How many ways can 3 identical red flags and 2 identical blue flags be arranged in a row?

7. How many arrangements of the letters of SPEED have the two E's together?

8. In how many ways can 6 people sit around a round table? (Rotations count as the same.)

9. True or false: the number of ways to seat 8 people around a round table equals the number of ways to arrange 7 people in a row.

10. How many arrangements are there of the letters of TATTY?

Section B Manipulation Drills ($\leq$50 s each)

Glue blocks, count gaps, divide by repeats. Name the technique before computing.

1. Six people stand in a row; Priya and Quinn stand together. How many arrangements?

2. Six people stand in a row; Priya and Quinn are *not* together. How many arrangements?

3. How many arrangements are there of the letters of MISSISSIPPI?

4. Five boys and three girls stand in a row so that no two girls are adjacent. How many arrangements?

5. Four men and four women stand in a row, alternating gender. How many arrangements?

6. Six people sit around a round table; two particular people insist on sitting together. How many seatings? (Rotations count as the same.)

7. How many six-digit strings use each of the digits $1, 1, 2, 2, 3, 3$?

8. How many arrangements are there of the letters of LEVEL?

9. Seven different books are shelved in a row. Three of them form a numbered trilogy which must appear in volume order (volume 1 before 2 before 3), though not necessarily adjacently. How many shelvings?

10. The number of ways to seat n people around a round table is 720. Find n.

Section C Substitution & Structure (≤90 s each)

Combine blocks, gaps and circles. Decide what is glued, what is free.

1. Eight people stand in a row with three particular friends in consecutive positions. How many arrangements?

2. Same eight people, the three friends consecutive *and* in height order (tallest first). How many arrangements now?

3. How many arrangements are there of the letters of DIVIDED?

4. Four men and four women sit around a round table, alternating gender. How many seatings? (Rotations count as the same.)

5. How many arrangements of the letters of CIRCLE have the two C's apart?

6. How many five-digit numbers use each of $1, 2, 3, 4, 5$ exactly once and are even?

7. Three maths books, two physics books and two chemistry books (all different) are shelved with books of the same subject together. How many shelvings?

8. Eight people sit around a round table; two particular people refuse to sit next to each other. How many seatings? (Rotations count as the same.)

Section D Challenge (SMC difficulty)

Insight over computation. If the last ones resist, sit with them — learn the tool, then return.

1. How many seven-digit numbers use each of $1, 2, \ldots, 7$ exactly once with the odd digits appearing in increasing order from left to right (not necessarily adjacent)?

2. Ten children stand in a circle. Mia and Noah must be adjacent, while Omar and Priya must *not* be adjacent. How many circles are possible? (Rotations count as the same.)

3. How many arrangements of AABBCC have no two identical letters adjacent?

4. Eight people, including three sisters, stand in a row so that *exactly* two of the sisters are adjacent. How many arrangements?

5. Eight knights sit at a round table. Three of them are feuding, and no two of the three may sit next to each other. How many seatings are possible? (Rotations count as the same.)

"Glue what must be together, seat what must be apart last — the gaps will tell you how much room rebellion really has."

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other prep to look at today:
Daily Integral for Calculus and Logic